

Julia Abboud

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EDUCATION

BS in Mechanical Engineering, University of Colorado Boulder

Graduation Date: May 2027

Minors: Engineering Management & Engineering Entrepreneurship

GPA: 3.989

Relevant Courses: Component Design, Heat Transfer, Thermodynamics, Fluids, Mechanics of Solids, Materials Science

Awards: Norlin Scholar, Esteemed Scholars Program, Dean's List

SKILLS

Technical: Digital Art (6 yrs), NX (750+ hrs), SolidWorks (CSWA, 220+ hrs), Lab Skills (220 hrs), Lathe/Mill (70+ hrs)

Programming: C++ (150 hrs), MATLAB (20 hrs), ANSYS (20 hrs), HTML/CSS (25 hrs), Arduino (10 hrs), EES (10 hrs)

Languages: Seal of Bilingualism in Spanish, Conversational Arabic

ENGINEERING EXPERIENCE

Apple Watch Product Design Intern, Cupertino, CA

Jan - Aug 2026

- Drove 5-10 mechanical design projects at a time for upcoming products in NX, including housing modifications and sheet metal cowlings, while managing vendor DFM feedback loops and drawing releases
- Led 5+ DOEs to validate the performance of snap-fits, watch band lug assemblies, and set 2 system reliability specs for the 2027 Watch using custom fixtures, heat soak environments, and FEA results
- Generated 10+ enclosure retention concepts and synthesized FEA, reliability data, and analytical stiffness modeling to deliver detailed risk assessments and actionable design recommendations
- Characterized materials with optical transmission and stress corrosion cracking studies to evaluate trade-offs

Drill-Powered Bike, CU Boulder

Sep - Dec 2025

- Served as project manager on a team of 5 to design a hand-drill-powered vehicle optimized for maneuverability
- Delivered a functional product 2 weeks ahead of schedule, 33% under-budget, and 4th place at a course-wide race
- Proactively resolved conflicts and filled in gaps, contributing heavily to system CAD, 20+ technical drawings, manufacturing, assembly, and drive chain alignment

Medtronic R&D Intern, Oximetry & Wearables, Boulder/Lafayette, CO

Jun - Aug 2025

- Prototyped a wearable sensor able to identify between 4 sites on the body based on curvature; evaluated strain gauges, flex sensors, and force sensors for performance through testing, calculations, and MATLAB analysis
- Validated sensor with a custom strain gauge amplifier, Wheatstone Bridge, Arduino, and flexible PCB circuit
- Created and 3D printed test fixtures and two 8-part assemblies in SolidWorks to accelerate team communication

Medtronic WISE R&D Intern, Oximetry & Wearables, Boulder, CO

Jun - Aug 2024

- Applied Systems Engineering principles to create a new requirements doc for an outdated analytics software by collecting 40+ stakeholder voices and drafting 10+ visual designs in preparation for contractor handoff
- Designed and executed laboratory ambient light testing for a next-generation pulse oximetry sensor, analyzed data to create new specs for use in a clinical environment
- Reworked a 50+ page ambient light test procedure to be 3x faster for future product validation and verification

LEADERSHIP EXPERIENCE

Machine Shop Assistant, CAD & Fabrication, CU Boulder

Feb 2025 - Jan 2026

- Instructed 50+ student groups to operate the mill and lathe safely and efficiently during 1-on-1 sessions
- Guided students through DFM principles while co-machining 6 distinct components of a wobbler engine

Teaching Assistant, Mechanical Engineering as a Profession, CU Boulder

Aug 2024 - Jan 2026

- Taught and modeled career skills such as elevator pitching, interviewing, and job applications for 60+ students
- Critiqued 30+ resumes, portfolios, and cover letters, supporting students to achieve career goals

STUDENT AFFILIATIONS

Buffy Buffs, Founder & President

Mar 2025 - Present

Taylor Swift Club, Vice President & Treasurer

Dec 2023 - May 2025